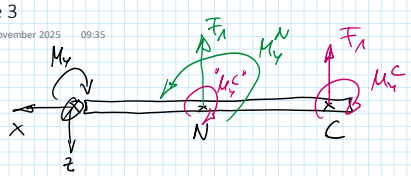
Exercise 3

Mittwoch, 5. November 2025 09:35

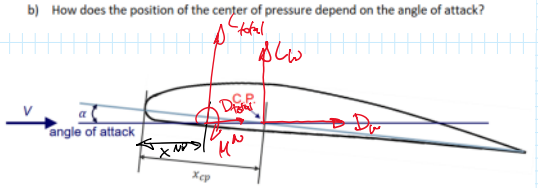


$$M_y^N = -F_A \cdot |x^{CW}| + M_y^C$$

Exercise 1: Neutral Point of the Wing

a) What holds true for the pitch moment around the neutral point?

b) How does the position of the center of pressure depend on the angle of attack?



a) $M^N \neq 0$

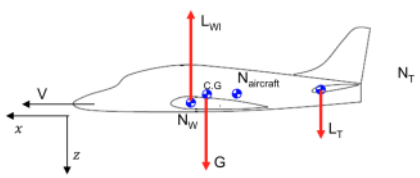
$\frac{\partial M^N}{\partial x} = 0$, $\frac{\partial x^N}{\partial x} = 0$

b) $\frac{\partial x^{cp}}{\partial x} \neq 0$, $M^{cp} = 0$

Exercise 2: Neutral Point of the Aircraft

In this exercise we will investigate the neutral point N of the aircraft. We assume that,

- the aircraft is in horizontal steady state cruise flight without wind
- both the wing and horizontal stabilizer have a purely horizontal inflow
- drag and thrust can be neglected



Lift of the wing	$\|L_W\|$	17000 N
Change of the wing lift with the angle of attack	$\Delta L_{W,\alpha}$	2000 N/°
Position of the neutral point of the wing N_W	x_{N_W}	−2.3m
Lift of the stabilizer (Tail)	$\|L_T\|$	1500 N
Change of the stabilizer lift with the angle of attack	$\Delta L_{T,\alpha}$	200 N/°
Position of the neutral point of the stabilizer N_T	x_{N_T}	−7m
Position C.G.	x_G	~~−1.8m~~ −1.8m

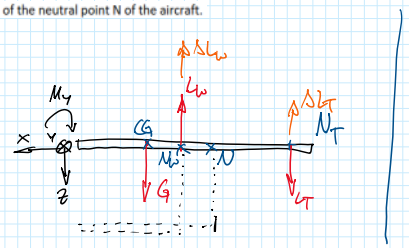
a) Calculate the mass of the aircraft.

$$\Sigma F \doteq 0 : \quad 0 = -L_W + L_T + G \rightarrow G = L_W - L_T$$

$$m \cdot g_0 = L_W - L_T$$

$$m = \frac{L_W - L_T}{g_0} = 1580\,kg$$

b) Calculate the position of the neutral point N of the aircraft.

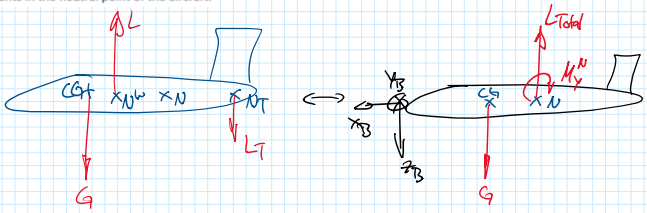


$$\Delta M^N = 0 : \quad \Delta L_W \cdot x^{NM_W} - \Delta L_T \cdot x^{NM_T} = 0$$

$$\Delta L_W \cdot \left(x^{M_W} - x^N\right) - \Delta L_T \left(x^N - x^{M_T}\right) = 0$$

$$x^N = \frac{\Delta L_W x^{M_W} + \Delta L_T \cdot x^{M_T}}{\Delta L_W + \Delta L_T} = -2.41\,m;$$

c) Calculate the moment M^N in the neutral point.
Hint: Assemble forces and moments in the neutral point of the aircraft.



$$\Sigma M_y^G = 0, \quad \Sigma F_z \doteq 0 \rightarrow L = G = 15530\,N;$$

$$0 = -L \cdot \left|x^{GN}\right| + M_y^N$$

$$M_y^N = L \cdot \left|x^{GN}\right| = 9455\,Nm;$$